SEQUENCE 12

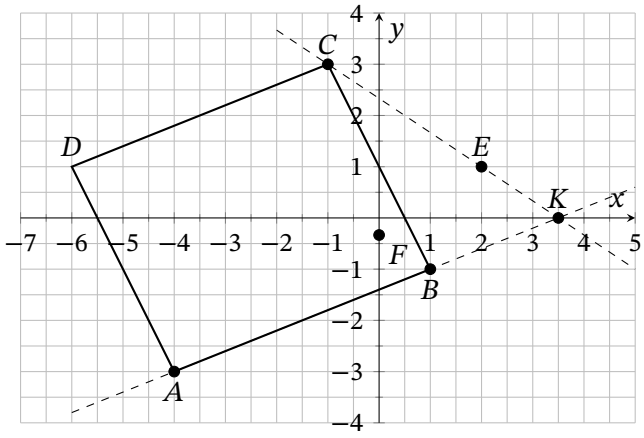
Coordinate geometry: vectors and lines synthesis

15–26 March 2027

12.1 Parallelogram and intersection

Example 1 · Booklet synthesis. In an orthonormal frame, $A(-4,-3)$, $B(1,-1)$ and $C(-1,3)$.

1. Determine D such that $ABCD$ is a parallelogram.
2. Determine its centre I in two ways.
3. With $E(2,1)$, determine K, the intersection of (AB) and (CE).
4. Determine F such that $\overrightarrow{AF} = \frac{2}{3}\overrightarrow{AE}$. Are D, F, B collinear?



Solution. 1. $D = A + C - B = (-6, 1)$. 2. Midpoint of $[AC]$ and of $[BD]$: $I(-5/2, 0)$. 3. (AB) : $2x - 5y - 7 = 0$; (CE) : $2x + 3y - 7 = 0$. Subtracting gives $8y = 0$, then $x = 7/2$, so $K(7/2, 0)$. 4. $\overrightarrow{AE} = (6, 4)$, so $F = A + (4, 8/3) = (0, -1/3)$. $\overrightarrow{DF} = (6, -4/3)$, $\overrightarrow{DB} = (7, -2)$; determinant $-12 + 28/3 = -8/3 \neq 0$. The three points are not collinear.

12.2 Comparing a vector method and a coordinate method

Example 2. In a non-degenerate triangle ABC, I is the midpoint of $[AB]$ and J the midpoint of $[AC]$.

1. Prove with vectors that $(IJ) \parallel (BC)$.
2. Check with $A(0,0)$, $B(6,0)$, $C(2,4)$.
3. For these coordinates, determine the equations of (IJ) and (BC), then the ratio IJ/BC.

Solution. 1. $\vec{IJ} = \vec{IA} + \vec{AJ} = \frac{1}{2}(\vec{BA} + \vec{AC}) = \frac{1}{2}\vec{BC}$. The vectors are nonzero and collinear.
2. $I(3,0)$, $J(1,2)$; $\vec{IJ} = (-2,2)$ and $\vec{BC} = (-4,4)$.
3. $(IJ) : x+y-3=0$; $(BC) : x+y-6=0$. Their direction vectors are collinear. $IJ = \sqrt{8}$, $BC = \sqrt{32}$, so $IJ/BC = 1/2$. The vector proof is general; the numerical check only proves this particular case.

Example 3. A figure seems to show two parallel lines. What is the difference between this observation and a proof?

Solution. The figure suggests a conjecture and may be imprecise. A proof uses exact data and a property, for example a zero determinant of nonzero direction vectors. A conclusion must state the objects concerned.

12.3 Recognising a configuration and justifying

Example 4 · Original situation. In an orthonormal frame, $A(0,0)$, $B(4,2)$, $C(3,4)$ and $D(-1,2)$.

1. Show that $ABCD$ is a parallelogram.
2. Calculate the four lengths and determine whether it is a rhombus.
3. Calculate AC^2, BD^2 and determine whether it is a rectangle.
4. Determine its centre and an equation of each diagonal.

Solution. 1. $\overrightarrow{AB} = (4,2) = \overrightarrow{DC}$: it is a parallelogram. 2. $AB = CD = \sqrt{20} = 2\sqrt{5}$, $BC = DA = \sqrt{5}$; adjacent sides are not equal, so it is not a rhombus. 3. $AC^2 = 3^2 + 4^2 = 25$; $BD^2 = (-5)^2 + 0^2 = 25$. A parallelogram whose diagonals have the same length is a rectangle. One can also use $AB^2 + BC^2 = 20 + 5 = 25 = AC^2$ and the converse of Pythagoras. 4. The centre is $I(3/2, 2)$. $(AC) : 4x - 3y = 0$; $(BD) : y = 2$. Their intersection recovers I.

Teacher note. Comparison of approaches: a proof by Pythagoras or by the property of the diagonals is enough. Do not impose the dot product, which is not needed here. For the oral part, distribute the methods among groups, then ask for a joint justification.